11. Acceptance-Rejection 3

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Given an $f(x)$, we choose an $h(x)$

and write $f(x) = \underset{\geq 1}{c}\, \underset{\in [0,1]}{g(x)}\, h(x)$

Start with $\dfrac{f(x)}{h(x)}$ and Maximize this to find c

Then $\dfrac{1}{c}\dfrac{f(x)}{h(x)} = g(x)$

* Assignment 1: $f(x)$ = Wigner's Spherical Law

$h(x)$ = the Beta distributions

Note: $f(x)$ is supported on $[-R, R]$

Beta$(\cdot,\cdot)$ are on $[0,1]$

Last time: Half Normal from Exp(1)

what if we chose a different $\lambda \neq 1$?

$$f(x) = \sqrt{\frac{2}{\pi}}\, e^{-x^2/2} \qquad h(x) = \lambda e^{-\lambda x}$$

$$\overset{\text{Max}}{\sup_{x \geq 0}} \left\{ \frac{f(x)}{h(x)} \right\} = \sup \left\{ \frac{1}{\lambda}\sqrt{\frac{2}{\pi}}\, e^{-x^2/2 + \lambda x} \right\}$$

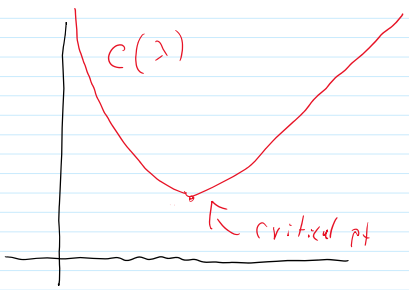
?

Try $\log$ and $\dfrac{d}{dx}$ and look for critical point

$$\log\left(\frac{f}{h}\right) = \log\sqrt{\frac{2}{\pi}} - \log\lambda - \frac{x^2}{2} + \lambda x$$

$$\frac{d}{dx}\log\left(\frac{f}{h}\right) = -x + \lambda, \qquad x = \lambda$$

$$c = \frac{1}{\lambda}\sqrt{\frac{2}{\pi}}\exp\left(-\frac{\lambda^2}{2} + \lambda^2\right) = \boxed{\frac{1}{\lambda}\sqrt{\frac{2}{\pi}}\, e^{\lambda^2/2}}$$



$$\log(c(\lambda)) = \log\sqrt{\frac{2}{\pi}} - \log\lambda + \lambda^2/2$$

$$\frac{d}{d\lambda}\log c(\lambda) = -\frac{1}{\lambda} + \lambda = 0 \Rightarrow \lambda^2 = 1, \quad |\lambda| = 1$$

as $\lambda \geq 0$, we get $\boxed{\lambda = 1}$

Maximize the $P(\text{Accept})$ over all $\lambda > 0$

- c is from maxing $\frac{f(x)}{h(x)}$ over x ← Def of c
- but if h depends on a parameter $\lambda > 0$ then we can minimize over λ to get the smallest c possible

* in the assignment you have to calculate c for $h(x)$ being Beta(1,1), Beta(1/2, 1/2), Beta(3/2, 3/2)

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## Acceptance-Rejection for half-normal
## using exponential lambda

ar.halfNorm <- function( lmb = 1, max.it = 100 ){
  for( ii in 1:max.it ){
    zz <- rexp( 1, lmb );  # Generate Z from h(z)
    uu <- runif(1, 0,1 );  # Generate U from Unif[0,1]
    if( uu < exp( -(zz - lmb)^2/2 ) ) # Accept or Reject
      break;
  }
  return(  # return a list object
    list(
      X  = zz, # random variate from f
      it = ii  # number of iterates
    )
  )
}
```

keep iterating until I get an "Accept"

$Z \sim h(x)$

$U \sim Unif$

$g(z)$

leave the for loop

is $\{U \le g(Z)\}$?

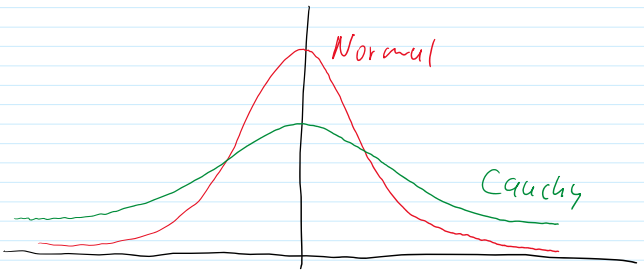
on average, this will take c iterations to stop

When we pick a distribution $h(x)$ to sample from we want $c \ge 1$ to be as small (i.e. close to 1) as possible.

Example: consider Normal vs Cauchy dist

$N(0,1)$: $\frac{1}{\sqrt{2\pi}} e^{-x^2/2}$

Cauchy: $\frac{1}{\pi} \frac{1}{1+x^2}$



if I want to pick one of these for $f(x)$ and one for $h(x)$, I would have to solve for c by maxing $\frac{f}{h}$ over all x

$$\frac{N(0,1)}{\text{Cauchy}} = c\,(1+x^2)\,e^{-x^2/2}$$ ← Does work ✓

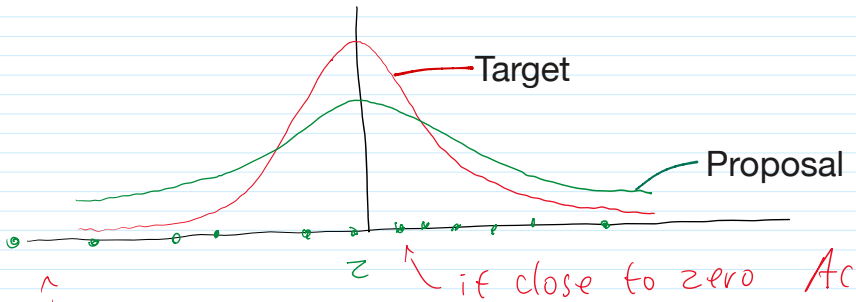
Why will it work?????

$$\frac{\text{Cauchy}}{} = c' \frac{}{e^{x^2/2}}$$ ← blow up to ∞

$$\frac{\text{Cauchy}}{N(0,1)} = C' \frac{e^{x^2/2}}{1+x^2}$$

blow up to ∞ as $x \to \infty$ or $x \to -\infty$

Max over x is ∞
thus can't do Accept / Reject



if close to zero Accept with Higher Probability

if far from zero, Accept with low Probability